

Experiment-15

Aim

To perform image transformation using the Direct Linear Transformation (DLT) algorithm by estimating a homography matrix from corresponding points and applying the transformation using OpenCV in Python.

Procedure

1. Import the required libraries (cv2 and numpy).
2. Load the input image.
3. Define four corresponding source points and destination points.
4. Compute the homography matrix using the Direct Linear Transformation (DLT) method through cv2.findHomography().
5. Apply the transformation using cv2.warpPerspective().
6. Display the original and transformed images.
7. Save the transformed image if required.

Code:

```
import cv2

import numpy as np

image = cv2.imread("sample.jpg")

if image is None:

    print("Error: Image not found!")

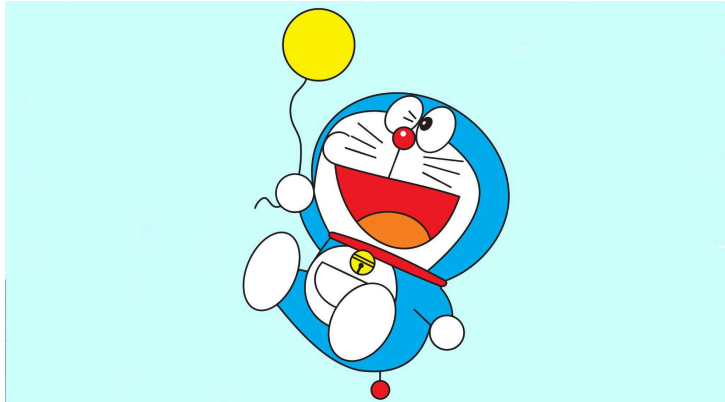
    exit()

src_points = np.array([

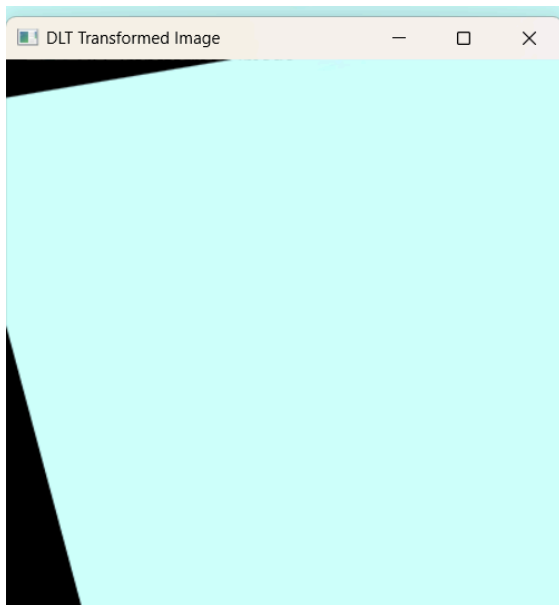
    [50, 50],
```

```
[300, 50],  
  
[50, 300],  
  
[300, 300]  
  
], dtype=np.float32)  
  
dst_points = np.array([  
  
[20, 80],  
  
[280, 40],  
  
[80, 320],  
  
[320, 300]  
  
], dtype=np.float32)  
  
H, mask = cv2.findHomography(src_points, dst_points, method=0)  
  
transformed = cv2.warpPerspective(image, H, (400, 400))  
  
cv2.imshow("Original Image", image)  
  
cv2.imshow("DLT Transformed Image", transformed)  
  
cv2.waitKey(0)  
  
cv2.destroyAllWindows()
```

Sample Input:



Output:



Results:

The image transformation using the Direct Linear Transformation (DLT) algorithm was successfully performed. The homography matrix was estimated from the corresponding source and destination points, and the transformed image was obtained using `cv2.warpPerspective()`.

